

Mukesh Ramdam

(817) 286-3052

mukeshramdam.dev@gmail.com

linkedin.com/in/mukesh-ramdam/

PROFESSIONAL SUMMARY

Innovative and detail-oriented **Senior Data Engineer** with **7+ years** of hands-on experience designing, developing, and deploying high-performance data platforms in both **cloud-native** and **hybrid environments**. Expertise spans healthcare and banking domains, with a strong focus on building scalable **ETL/ELT workflows**, **real-time streaming solutions**, and **data lakehouse architectures**.

Proficient in **Azure** (ADF, Synapse, ADLS Gen2, SQL DB, Key Vault, Purview) and **AWS** (Glue, Redshift, EMR, Lambda, S3, IAM), leveraging these platforms to deliver end-to-end data pipelines and integrate cloud solutions seamlessly. Proven track record in modernizing legacy ecosystems and transitioning to next-generation platforms such as **Snowflake** and **Azure Synapse**.

Experienced in **Apache Spark** (PySpark, Scala), **Kafka**, **Airflow**, and **Databricks** for data processing and orchestration. Skilled in **data modeling**, **SQL optimization**, and building **analytical data marts** using tools like **Erwin**. Delivered real-time **clinical analytics** using **FHIR APIs**, **Spark Streaming**, and integrated **Power BI/Tableau** dashboards, enriched with **DAX** and **Power Apps**.

Applied **ML/AI** techniques including **scikit-learn**, **XGBoost**, **TensorFlow**, and **Keras** for predictive analytics in healthcare workflows, including LLM-powered automation of unstructured data. Led infrastructure automation using **Terraform**, **Docker**, **Kubernetes**, and streamlined **DevOps pipelines** with **Jenkins**, **GitLab**, and **GitHub Actions**.

Adept in ensuring data privacy and regulatory compliance (HIPAA, **GDPR**, **SOC2**) through secure and governed data pipeline designs. Proficient in **Python**, **Bash**, **SQL**, **Hadoop** (HDFS, Hive, HBase), **MongoDB**, **Oracle**, **Redshift**, and **multi-format ingestion** (Parquet, Avro, XML).

TECHNICAL SKILLS

- **Cloud Platforms:** Azure (Data Factory, ADLS Gen2, Azure Databricks, Synapse Analytics, Azure SQL Database, Microsoft Fabric, Key Vault, Purview, Azure DevOps), AWS (Amazon Redshift, S3, Glue, Lambda, IAM, CloudFormation), GCP (BigQuery)
- **Data Engineering & Orchestration:** Apache Spark (PySpark, Scala), Apache Kafka, Airflow, Databricks, Azure Data Factory, AWS Step Functions, AWS Lambda
- **Data Warehousing & Data Lakes:** Delta Lake, Azure Synapse Analytics, Snowflake, Amazon Redshift, AWS Lake Formation, Azure Data Lake Storage Gen2
- **Machine Learning & AI:** scikit-learn, XGBoost, TensorFlow, Keras, Spark MLlib, MLflow, Predictive Modeling, LLM-powered Automation
- **ETL/ELT Tools:** Fivetran, dbt, Informatica, Talend, Apache NiFi, AWS Glue, Azure Data Factory, Apache Kafka, Apache Sqoop
- **DevOps & Automation:** Terraform, Docker, Kubernetes, Jenkins, GitLab, GitHub Actions, Azure DevOps, CI/CD Pipelines

- **Data Modeling & SQL Optimization:** Erwin, Data Vault, Star Schema, Snowflake Schema, SQL Query Optimization, Query Performance Tuning
- **Data Visualization & Reporting:** Power BI, Tableau, DAX, Power Apps, Looker, Data Dashboards, Real-time Analytics
- **Big Data & NoSQL:** Hadoop (HDFS, Hive, HBase), MongoDB, Parquet, Avro, Sequence Files, JSON, XML
- **Data Privacy & Compliance:** HIPAA, SOC2, GDPR, Data Governance, Encryption, Data Lineage, Azure Key Vault, AWS KMS
- **Programming & Scripting:** Python, SQL, Bash, Java, Scala, Spark SQL

PROFESSIONAL EXPERIENCE

Pfizer - South San Francisco, CA

Senior Data Engineer | May 2022 – Present

- Architected enterprise-grade Azure lakehouse and warehouse solutions for clinical, R&D, pharmacovigilance, and manufacturing analytics by combining **Azure Data Factory**, **ADLS Gen2**, **Azure Databricks**, **Synapse Analytics**, and **Azure SQL Database** to deliver scalable ingestion, transformation, and curated reporting layers aligned with client business objectives.
- Designed and optimized complex dimensional models and **star schemas** for regulated healthcare reporting, creating conformed dimensions, fact tables, surrogate key strategies, and semantic-ready data marts that improved downstream analytics performance and reporting consistency across clinical and operational domains.
- Built metadata-driven ELT frameworks using **Azure Data Factory** and **Python** to orchestrate reusable ingestion patterns, parameterized pipelines, dynamic schema handling, exception workflows, dependency management, and automated notifications for reliable enterprise-scale data delivery.
- Developed advanced transformation logic in **SQL** and **PySpark** to standardize, flatten, and normalize deeply nested JSON payloads from FHIR, CTMS, EDC, LIMS, MES, and external API sources into analytics-ready relational structures for downstream dimensional modeling and operational reporting.
- Implemented secure API ingestion patterns using **OAuth 2.0**, REST connectors, Azure secrets management, and tokenized authentication workflows to integrate third-party and internal healthcare application data into governed Azure data platforms with full traceability and controlled access.
- Optimized large-scale transformation workloads on **Azure Databricks** and **Spark** for high-volume clinical and scientific datasets, improving pipeline throughput, reducing processing latency, and supporting near-real-time analytics use cases for safety monitoring and manufacturing quality insights.
- Partnered with BI developers, analysts, cloud administrators, and business stakeholders to run architecture workshops, define target-state data strategies, document solution diagrams, and translate reporting and integration requirements into repeatable Azure data engineering patterns.
- Established best-practice data quality frameworks with automated validations, reconciliation checks, schema drift handling, audit columns, and pipeline observability using **Python**, **SQL**, and orchestration controls to improve reliability, modularity, and supportability of enterprise data products.
- Applied modern lakehouse architecture principles by integrating raw, standardized, and curated zones across **ADLS Gen2**, **Synapse Analytics**, and **Microsoft Fabric**, enabling governed analytics, scalable storage, and strong separation between ingestion, transformation, and consumption layers.
- Strengthened security and governance with **Azure Key Vault**, **Purview**, role-based access controls, lineage capture, data classification, and encryption controls to support HIPAA, GDPR, and GxP aligned data management for sensitive healthcare datasets.
- Created reusable transformation models and reporting-ready marts with **dbt** and **SQL** patterns where appropriate for modular business logic, testing, documentation, and clearer handoff between engineering and analytics teams.
- Contributed cross-cloud integration patterns by publishing curated Azure datasets to **Amazon Redshift** and coordinating data exchange with **AWS Glue** and **S3** where client ecosystems required hybrid reporting or migration support.

Technologies: Azure Data Factory, ADLS Gen2, Azure Databricks, Spark, Synapse Analytics, Azure SQL Database, Microsoft Fabric, Python, SQL, dbt, Azure Key Vault, Purview, Azure DevOps, Power BI, Amazon Redshift, AWS Glue, S3, Lambda, IAM

Optum - New York, NY

Data Engineer | Jan 2020 - Mar 2022

- Designed and developed scalable healthcare data pipelines on AWS and Azure to ingest claims, eligibility, provider, member, and operational datasets, while aligning transformations and reporting structures with enterprise analytics and dimensional modeling requirements.
- Built reusable ingestion and orchestration workflows using **Python, SQL, AWS Glue, and Azure Data Factory**, enabling metadata-driven processing, standardized load patterns, automated dependency handling, and consistent execution across batch and near-real-time data pipelines.
- Modeled healthcare reporting datasets using dimensional design techniques and **star schemas**, creating fact and dimension structures for claims analytics, utilization reporting, cost of care analysis, and operational KPI dashboards.
- Developed complex **SQL** transformations and performance-tuned warehouse logic in **Amazon Redshift** to support analytical workloads, scalable reporting, and downstream business intelligence consumption across payer and operations teams.
- Engineered secure ingestion from APIs and semi-structured healthcare sources by parsing, flattening, and normalizing nested JSON into relational warehouse structures, using **Python** and SQL-based transformation rules to improve downstream usability and data consistency.
- Built governed data lake patterns on **Amazon S3** with curated transformation layers, partitioning standards, access controls, and downstream warehouse publishing to support reliable analytics and cost-effective storage at scale.
- Implemented data quality controls, reconciliation rules, exception handling, and operational alerting across ingestion and transformation pipelines to improve reliability, reduce manual intervention, and strengthen trust in analytics outputs.
- Collaborated with analysts, reporting teams, and platform stakeholders to align business requirements with scalable engineering solutions, translating reporting needs into curated warehouse tables, documented transformation logic, and reusable pipeline components.
- Supported modular data transformation practices with **dbt**, including tested SQL models, reusable logic, and clearer separation of staging, intermediate, and mart layers for maintainable analytics engineering workflows.
- Integrated event-driven and serverless processing using **Lambda** and orchestration services to automate file arrivals, trigger downstream transformations, and support responsive data delivery across operational healthcare systems.
- Maintained secure cloud data environments with **IAM**, encryption controls, access policies, and audit-ready operational practices to protect PHI and support HIPAA-aligned engineering standards.

Technologies: Amazon Redshift, SQL, Python, dbt, AWS Glue, S3, Lambda, IAM, Athena, Lake Formation, Azure Data Factory, Azure Key Vault, Terraform, QuickSight

Etsy - Brooklyn, NY

Big Data Developer | Sep 2018 - Nov 2019

- Designed large-scale data processing systems using **Hadoop MapReduce**, processing high-volume **e-commerce data** to support analytics for marketing, sales, inventory, and supply chain teams.
- Built and maintained scalable batch ETL pipelines using **Apache Pig** and **Apache Hive**, transforming raw and semi-structured data into analytics-ready datasets on **HDFS**.
- Developed and optimized SQL-based data models and queries in **Apache Hive**, enabling efficient analytical reporting for customer behavior analysis and inventory management.
- Implemented custom **MapReduce jobs in Java** to support complex transformations such as product categorization, sales trend analysis, and customer segmentation.

- Integrated **relational databases (Oracle, MySQL)** with the **Hadoop ecosystem**, enabling seamless ingestion of structured data for unified analytics.
- Automated ingestion of log, clickstream, and event data using **Apache Flume** and orchestrated batch workflows with **Apache Oozie**, ensuring reliable and timely pipeline execution.
- Implemented real-time data streaming using **Apache Kafka**, enabling near real-time insights into transactional data and user behavior.
- Monitored and maintained **Hadoop clusters** using **Apache Ambari and Ganglia**, ensuring high availability, performance tuning, and efficient resource utilization.
- Collaborated with data science and analytics teams to support machine learning and advanced analytics **use cases**, including recommendation systems and customer segmentation, using **Apache Mahout** and curated Hive datasets.
- Designed fault-tolerant data ingestion and processing workflows leveraging **HDFS replication** and **MapReduce resiliency**, ensuring reliable processing of large-scale e-commerce datasets.

Technologies: Hadoop MapReduce, Apache Pig, Apache Hive, Java, Apache Flume, Apache Oozie, Apache Kafka, Apache Mahout, SQL Server Reporting Services (SSRS), Oracle, MySQL, Ganglia, Apache Ambari, HDFS, JasperReports, Shell scripting.

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- AWS Certified Data Engineer Associate (DEA-C01)
- Microsoft Certified: Azure Fundamentals (AZ-900)

EDUCATION

Master of Science in Computer Science

San Francisco State University - San Francisco, CA